3.31 VESTIGIAL SIDEBAND TRANSMISSION (VSB)*

(U.P. Tech, Sem. Exam; 2007-08) (10 marks)

The stringent frequency-response requirements on the sideband filter in SSB-SC system can be relaxed by allowing a part of the unwanted sideband (called as Vestige) to appear in the output of the modulator. Due to this, the design of the sideband filter is simplified to a great extent. But the bandwidth of the system is increased slightly. To generate a VSB signal, we have to first generate a DSB-SC signal and then pass it through a sideband filter as shown in figure 3.48. This filter will pass the wanted sideband as it is alongwith a part of unwanted sideband.

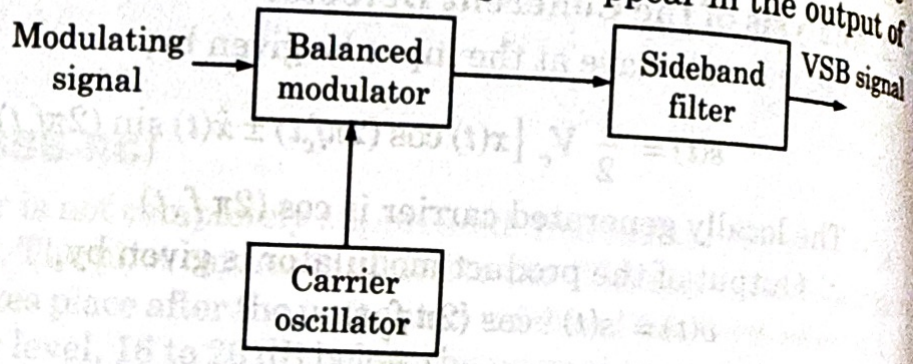

FIGURE 3.48 Generation of VSB signal.

3.31.1. Frequency Domain Description

1. Frequency Spectrum

The spectrum of VSB is as shown in figure 3.49. The spectrum of message signal $x(t)$ has also been shown. In the frequency spectrum, it is assumed that the upper sideband is transmitted as it is and the lower sideband is modified into vestigial sideband.

2. Transmission bandwidth

From figure 3.49(*b*), it is evident that the transmission bandwidth of the VSB modulated wave is given by,

$$B = (f_m + f_v) \text{ Hz}$$

where f_m = Message bandwidth and f_v = Width of the vestigial sideband.

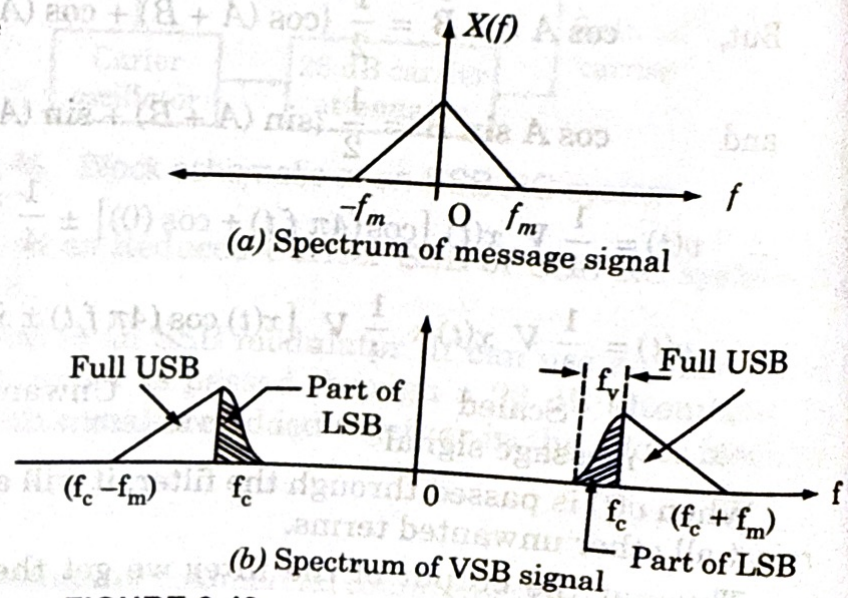

FIGURE 3.49

* Discuss how the VSB modulation is used in commercial TV signal. Discuss its merit and demerit

3. Advantages of VSB

(i) The main advantage of VSB modulation is the reduction in bandwidth. It is almost as efficient as the SSB.

(ii) Due to allowance of transmitting a part of lower sideband, the constraint on the filters have been relaxed. So practically, easy to design filters can be used.

(iii) It possesses good phase characteristics and makes the transmission of low frequency components possible.

4. Applications of VSB

VSB modulation has become standard for the transmission of Television signals. Because the video signals need a large transmission bandwidth if transmitted using DSB-FC or DSF-SC techniques.

3.31.2. Generation of VSB Modulated Wave

The block diagram of a VSB modulator is shown in figure 3.50.

The modulating signal $x(t)$ is applied to a product modulator. The output of the carrier oscillator is also applied to the other input of the product modulator. The output of product modulator is given by

$$m(t) = x(t) \cdot c(t) = x(t) \cdot V_c \cos(2\pi f_c t)$$

This represents a DSB-SC modulated wave. This DSB-SC signal is then applied to a sideband shaping filter. The design of this filter depends on the desired spectrum of the VSB modulated signal. This filter will pass the wanted sideband as it is and the vestige of the unwanted sideband. Let the transfer function of the filter be $H(f)$. Hence, the spectrum of the VSB modulated signalis given by,

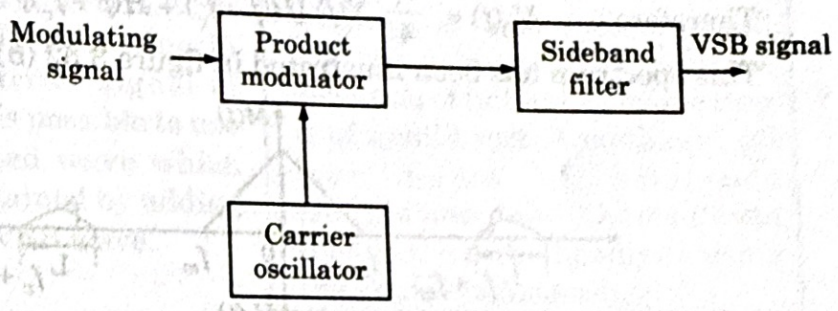

FIGURE 3.50 Generation of VSB signal.

$$S(f) = \frac{V_c}{2}\left[X(f - f_c) + X(f + f_c)\right] H(f) \quad \text{... (3.137)}$$

3.31.3. Demodulatiion or Detection of VSB Wave

1. Block Diagram

The synchronous detector for the detection of VSB modulated wave is shown in figure 3.51.

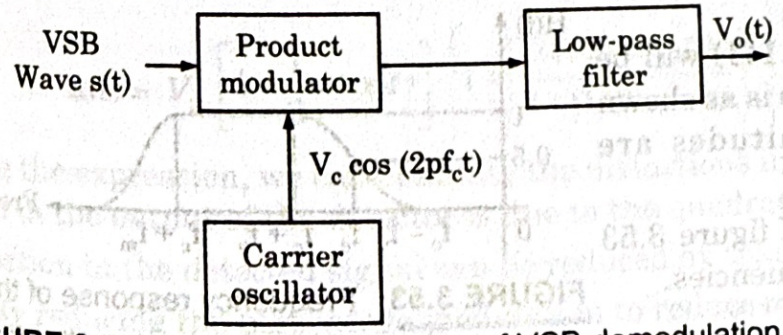

FIGURE 3.51 Synchronous detector of VSB demodulation.

DO YOU KNOW ?

VSB modulation requires a channel bandwidth that is between that required for SSB and DSB-SC systems, and the saving in bandwidth can be significant if modulating signals with large bandwidths are being handled, as in the case of television signals and high-speed data.

2. Working Operation

The VSB modulated wave is passed through a product modulator where it is multiplied with the locally generated synchronous carrier. Hence, the output of the product modulator is given by,

$$m(t) = s(t) \times c(t) = s(t) \cos(2\pi f_c t)$$

Taking the Fourier transform of both sides, we get

$$M(f) = S(f) * \left[\frac{1}{2}\delta(f+f_c)+\frac{1}{2}\delta(f-f_c)\right] = \frac{1}{2}S(f+f_c) + \frac{1}{2}S(f-f_c) \quad \text{...(3.138)}$$

But $$S(f) = \frac{V_c}{2}[X(f-f_c)+X(f+f_c)]\,H(f)$$

Hence, we have

$$M(f) = \frac{V_c}{4}[X(f-2f_c)\,H(f-f_c) + X(f+2f_c)\,H(f+f_c)] + \frac{V_c}{4}X(f)\,[H(f-f_c)\,H(f+f_c)] \quad \text{...(3.139)}$$

3. Frequency Spectrum

The spectrum $M(f)$ is illustrated in figure 3.52 (*a*). The first term in the above expression represents the VSB modulated wave, corresponding to a carrier frequency of $2f_c$. This term will be eliminated by the filter to produce output $v_0(t)$. The second term in equation (3.139) of $M(f)$ represents the spectrum of demodulated VSB output.

Therefore, $$V_0(f) = \frac{V_c}{4}X(f)\,[H(f-f_c)+H(f+f_c)] \quad \text{...(3.140)}$$

This spectrum has been illustrated in figure 3.52 (*b*).

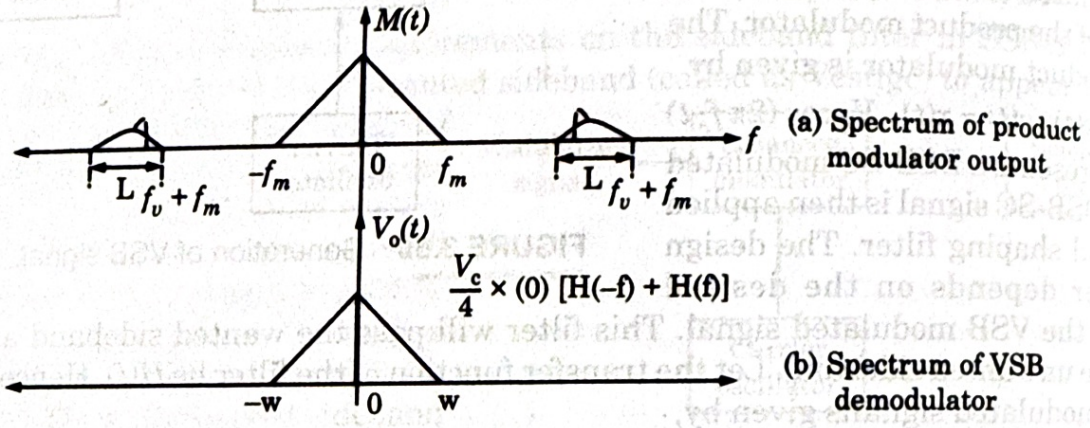

FIGURE 3.52

If we want to obtain the undistorted message signal $x(t)$ at the output of the demodulator, then $V_0(f)$ should be a scaled version of $X(f)$. For this the transfer function $H(f)$ should satisfy the condition as follows :

$$H(f-f_c) + H(f+f_c) = 2\,H(f+f_c) \quad \text{...(3.141)}$$

where $H(f_c)$ is constant.

The condition stated in equation (3.141) will be satisfied if the filter frequencies response is as shown in figure 3.53. Note that the magnitudes are normalized.

Note that the frequency response of figure 3.53 has been drawn only for the positive frequencies.

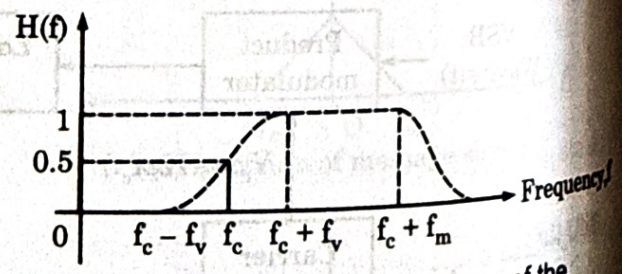

FIGURE 3.53 Frequency response of the sideband shaping filter for obtaining the VSB modulated wave with vestige of LSB.

4. Observation from figure 3.53.

Some important observations from figure 3.53 are as follows :

(*i*) As the frequency response is normalized, $H(f)$ reduces to 0.5 at $f = f_c$.

(*ii*) The cutoff portion of this frequency response, around f_c (from $f_c - f_v$ to $f + f_v$) exhibits an odd symmetry because in this range of frequency, the sum of $H(f)$ at two frequencies equally spaced above and below f_c will be unity.

(*iii*) The design of VSB filter is therefore less complicated as compared to an SSB filter.

Table 3.2. Comparison of A.M. Techniques

S.N.	Parameter of comparison	DSBFC (Standard AM)	DSBSC	SSB	VSB
(*i*)	Carrier suppression	N.A.	Fully	Fully	N.A.
(*ii*)	Sideband suppression	N.A.	N.A.	One S.B. completely	One S.B. suppressed partially
(*iii*)	Bandwidth	$2fm$	$2f_m$	f_m	$f_m < BW < 2f_m$
(*iv*)	Transmission efficiency	Minimum	Moderate	Maximum	Moderate
(*v*)	No. of modulating inputs	1	1	1	2
(*vi*)	Application	Radio broadcasting	Radio broadcasting	Point to point mobile communication	T.V.